

Movie Recommender System Tutorial using Collaborative Filtering

1 Introduction

A recommender system suggests items such as movies, books, or music based on user preferences. In this tutorial, a *Collaborative Filtering Recommender* that uses users' historical rating patterns to recommend movies is discussed.

This system reads movie titles from `Movie_Id_Titles_clean.csv`, which contains:

- **item_id**: unique movie identifier
- **title**: movie title

The user rating behavior is stored in `u.data`, which contains:

- **user_id**
- **item_id**
- **rating**
- **timestamp**

2 Collaborative Filtering

Collaborative Filtering (CF) recommends movies by analyzing patterns in how groups of users rate movies.

The key assumption is:

“If two movies received similar ratings from many users, they are likely to be similar and can be recommended together.”

We use *Item-Item Collaborative Filtering* where the system identifies movies with similar rating patterns.

3 Data Preparation

3.1 Loading Data

```
ratings = pd.read_csv("u.data", sep="\t",
                      names=["user_id", "item_id", "rating", "timestamp"])
```

```
movies = pd.read_csv("Movie_Id_Titles_clean.csv")
```

```
df = pd.merge(ratings, movies, on="item_id")
```

We merge ratings with movie titles using *item_id*.

3.2 Creating Rating Matrix

```
moviemat = df.pivot_table(index="user_id",
                           columns="title",
                           values="rating")
```

This produces a matrix of shape:

Users $\times$ Movies

where each cell is a rating.

4 Computing Similarity

To recommend movies similar to a selected movie, we compute the *Pearson correlation* between its rating vector and the rating vectors of other movies:

$$\text{corr}(A, B) = \frac{\sum (A_i - \bar{A})(B_i - \bar{B})}{\sqrt{\sum (A_i - \bar{A})^2 \sum (B_i - \bar{B})^2}}$$

Movies with high correlation scores are considered similar.

5 Filtering for Reliable Recommendations

Not all movies have enough ratings to compute meaningful similarity. Therefore, we only keep movies with a minimum number of ratings (for example, 25):

```
corr_df = corr_df[corr_df["num_ratings"] >= 25]
```

This improves result quality.

6 Example Output

Enter movie name: Toy Story

Matched: Toy Story (1995)

Top Recommended Movies:

	Correlation
Liar Liar (1997)	0.72
Jerry Maguire (1996)	0.68
Mrs. Doubtfire (1993)	0.64
Aladdin (1992)	0.61

7 Key Concepts

- This recommender does *not use genres or movie descriptions*.
- It relies entirely on *user rating similarity patterns*.
- Recommendations reflect *collective audience behavior*.

8 Collaborative Filtering vs Content-Based

<i>Content-Based Filtering</i>	<i>Collaborative Filtering</i>
Uses movie features (e.g., genre)	Uses user rating patterns
Recommends similar movies	Recommends movies liked by similar users
Works with few users	Needs many users and ratings

9 Conclusion

Collaborative Filtering is simple, interpretable, and effective for datasets with rich rating histories. It allows us to build practical recommenders without machine learning model training.